

Design of a microfluidic chip for gradient generation

ME 224, Fundamentals of Microscale Flows

Submitted by

REENA MEENA – 230003057

ROHIT RANJAN – 230003059

RUCHA JATIN PRABHU - 230003060

Course Coordinator:

Dr. Vijai Laxmi, Assistant Professor, ME



Department of Mechanical Engineering
Indian Institute of Technology, Indore

Abstract

Microfluidic gradient generators are essential for precisely controlling chemical concentration distributions at the microscale, enabling applications in drug delivery, biological assays, and tissue engineering. This study focuses on the design, simulation, and experimental validation of a microfluidic gradient generator utilizing a tree-like network combined with serpentine channels to achieve stable and reproducible gradients. Computational simulations were conducted to optimize channel geometry and flow conditions, with a simulated flow rate of 5.4×10^{-5} mL/min ensuring diffusion-driven mixing. However, experimental results at a higher flow rate of 0.2 mL/min demonstrated convection-dominated mixing, leading to transient gradient formation. The absence of a dedicated waste outlet further contributed to unintended flow disturbances. Despite these challenges, momentary gradient formation was observed, confirming partial functionality. Future work will focus on optimizing flow rates, refining channel geometry, and integrating real-time monitoring sensors to enhance gradient stability and reproducibility.

Content

1. Introduction.....	
2. Literature Review.....	
3. Motivation and Objective	
4. Methodology.....	
5. Results and Discussion.....	
6. Conclusions.....	
7. Future scope.....	

References

1. Introduction

Microfluidics is the study and application of fluid flow at the microscale, enabling precise control over small fluid volumes. This technology has gained significant attention due to its ability to manipulate fluids at the micrometer scale, offering advantages such as low reagent consumption, rapid processing times, and enhanced control over chemical and biological interactions.

One crucial application of microfluidics is gradient generation, which refers to the controlled formation of spatial variations in the concentration of a substance within a fluidic system. Gradients play a fundamental role in biological and chemical processes, particularly in cell signaling, drug testing, and tissue engineering.

Traditional gradient generation methods often rely on macroscale diffusion chambers or serial dilutions, which lack precision and repeatability. In contrast, microfluidic gradient generators offer higher accuracy and reproducibility by leveraging laminar flow conditions and controlled diffusion. These systems typically use **Y-shaped channels**, and **serpentine channels** (as depicted in Fig 1 and Fig 2) to achieve well-defined gradients.

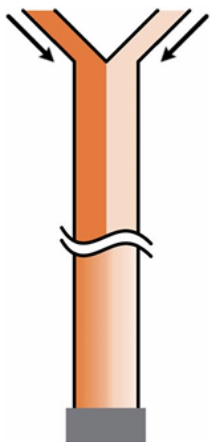


Fig 1: Y-shaped channel



Fig 2: Serpentine channel

2. Literature Review

Microfluidic gradient generators play a crucial role in creating precisely controlled chemical environments at the microscale. They enable detailed studies of processes by leveraging laminar flow, diffusion, and efficient mixing within microchannels. Achieving optimal performance requires careful design considerations to ensure stable and reproducible gradient formation.

2.1 Gradient Generation Methods and Design Principles

Wang, Liu, and Pang (2017) provide a comprehensive overview of concentration gradient generation methods in microfluidic systems. These methods include Y-shaped channels, tree-like networks(as shown in Fig 3), and serpentine channels, each offering distinct advantages in controlling fluid mixing. Our design strategically integrates a tree-like structure with serpentine channels to enhance mixing efficiency and ensure gradient stability.

Furthermore, the study highlights key design parameters influencing gradient formation, such as channel geometry, flow rate, and the diffusion coefficient of the involved substances. These factors collectively determine the stability and uniformity of the resulting gradient profile.

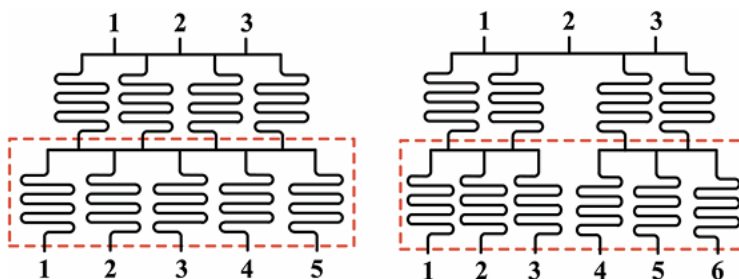


Fig 3: Tree like network

2.2 Fabrication Strategies and Applications

Pattanayak, Singh, and Gulati (2021) discuss recent advancements in microfluidic chip fabrication, particularly emphasizing soft lithography using polydimethylsiloxane (PDMS). This method is widely employed due to its cost-effectiveness, ease of fabrication, and biocompatibility, making it a preferred choice for rapid prototyping. Our microfluidic chip was fabricated using soft lithography, ensuring reliable and reproducible channel structures.

Regarding specific design considerations, Wang, Liu, and Pang (2017) review various gradient generation approaches but do not provide precise values for channel dimensions or flow rates. However, guidelines from the American Institute of Chemical Engineers (AIChE J, 2006) suggest that suitable channel dimensions range from 10 to 300 μm . The parameters used in our device were optimized through simulations to minimize gradient distortion.

3. Motivation and Objective

3.1 Motivation

Microfluidic gradient generators play a crucial role in biological studies and cell response analysis by enabling precise control over chemical and biological gradients at the microscale. Traditional designs often face challenges such as uneven mixing, dead zones, and high pressure drops, which can compromise gradient stability and experimental accuracy.

To address these limitations, our design incorporates an altered tree-shaped network that enhances gradient formation efficiency while maintaining low pressure drops. This design mimics natural biological conditions, making it well-suited for in vitro studies requiring controlled concentration profiles. Additionally, microfluidics significantly reduces reagent consumption, making experiments more cost-effective and sustainable. By improving flow control, stability, and mixing efficiency, our microfluidic chip provides a reliable platform for applications in drug testing, chemical analysis, and biomedical research.

3.2 Objectives

The primary objective of this study is to develop a highly efficient and adaptable microfluidic gradient generator that overcomes the limitations of existing designs. This involves:

- **Enhance stability and control** through a tree-like mixing network and optimized hydraulic resistance.
- **Minimize shear stress** by maintaining smooth channels and low flow rates.
- **Improve versatility** with adjustable inlet flow rates for various concentration profiles.
- **Ensure cost-effectiveness** by utilizing PDMS/glass for fabrication, ensuring reusability.
- **Enable high throughput** by using parallelized channels for multiple simultaneous gradient experiments.

4. Methodology

4.1 CAD Design & Mold Preparation

Designed the microfluidic chip in Fusion 360 with a tree-like network and serpentine channels. Mold was printed using a high-precision 3D printer. Chip dimensions: Outer box ($4.3\text{ cm} \times 2.6\text{ cm} \times 0.5\text{ cm}$), Inner chip ($3.5\text{ cm} \times 1.6\text{ cm}$), Hydraulic diameter (0.3 mm).

4.2 PDMS Casting & Solidification

Mixed PDMS with a curing agent in a 10:1 ratio. Degassed the mixture in a vacuum chamber. After that poured into the mold and thermally cured at 60°C for about 4 hours. Peeled off the solidified PDMS (Fig 4) for further processing.

4.3 Characterization & Bonding

Microchannel dimensions were verified using a digital microscope thereby confirming structural integrity before bonding the PDMS to a glass substrate.

4.4 Experimental Setup

The microfluidic chip was connected to a syringe pump. Two syringes with dye solutions were used to establish a concentration gradient. The gradient formation was analyzed under a DinoCapture microscope. The entire setup is shown in Fig 5.

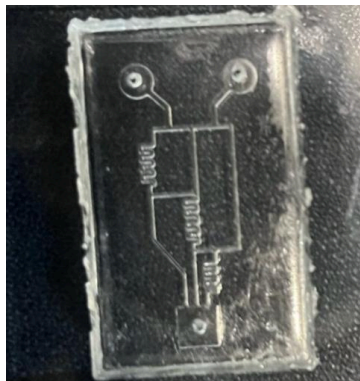


Fig 4: After Peeling

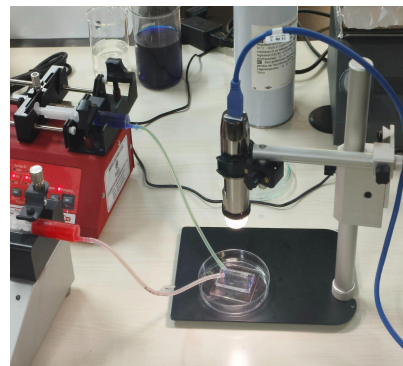


Fig 5: Setup

5. Results and Discussion

5.1 Theoretical Calculations

To determine the flow characteristics within the microfluidic gradient generator, we first estimated the Reynolds number (Re) to confirm the presence of laminar flow. The Reynolds number is given by:

$$Re = \frac{\rho U D}{\mu}$$

where ρ is the fluid density, U is the velocity, D is the characteristic length (hydraulic diameter), and μ is the dynamic viscosity. Given the small channel dimensions and low velocity, the calculated Reynolds number came out to be **0.00299**. This is significantly below 2000, verifying that the flow remained in the laminar regime.

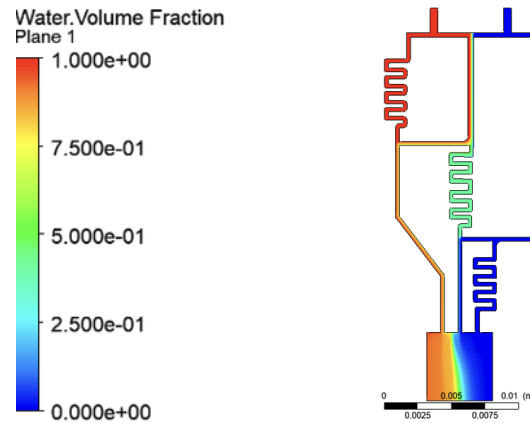
Additionally, the Peclet number (Pe) was calculated to assess the relative influence of advection and diffusion in the mixing process. The Peclet number is given by:

$$Pe = \frac{UL}{D_m}$$

where D_m is the molecular diffusion coefficient of the solute. The obtained Pe value was **30**, suggesting that both diffusion and advection played a role in solute transport. While advection was present, diffusion remained a significant factor in the formation of the concentration gradient. This highlights the importance of optimizing flow rates and channel dimensions to achieve the desired gradient profile.

5.2 Simulation and Analysis

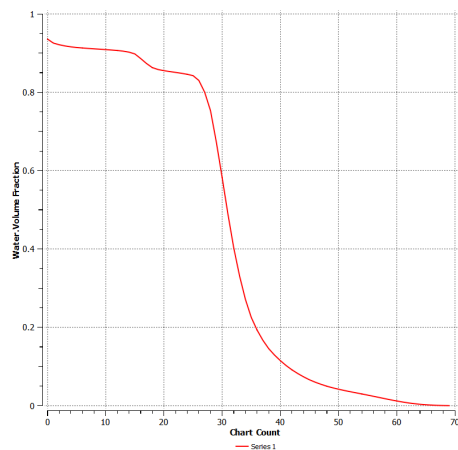
- Flow simulation performed in ANSYS Fluent using an Eulerian multiphase model.
- Inlet velocity: 10^{-5} m/s, ensuring steady-state flow.
- Velocity profiles analyzed to optimize channel design and improve mixing efficiency.



5.3 Water Volume Fraction Profile

The water volume fraction profile at the outlet exhibits anomalous diffusion characteristics, deviating from classical diffusion behavior.

- **Sharp Transition & Non-Gaussian Profile:** The profile shows a steep drop in water volume fraction, indicating a **sharp interface rather than a smooth gradient**, which is uncommon in purely diffusive systems.
- **Deviation from Fickian Diffusion:** Unlike normal Fickian diffusion, where the concentration profile follows a symmetric Gaussian distribution, the observed profile exhibits an **asymmetric tail**, suggesting non-Fickian transport mechanisms.



5.4 Hydraulic Resistance

The hydraulic resistance R_h was determined using both analytical and numerical approaches to validate the accuracy of the design.

For the analytical approach, we employed the theoretical formula:

$$R_{hyd} \approx \frac{28\mu L}{h^4}$$

where μ is the dynamic viscosity, L is the channel length, and h is the channel height. This expression is derived under the assumption of **fully developed, steady, and laminar flow** in a rectangular microchannel. It also assumes that the aspect ratio is sufficiently small for higher-order effects to be negligible. Additionally, the formula is an approximation because it does not account for entrance effects or variations in flow velocity across the channel width. The calculated hydraulic resistance using this formula was approximately **$5.91 \times 10^{10} \text{ Pa}\cdot\text{s}/\text{m}^3$** .

In the numerical approach, the hydraulic resistance was computed based on the pressure drop (ΔP) and flow rate (Q), obtained from ANSYS simulations, using the relation:

$$R_{hyd} = \frac{\Delta P}{Q}$$

Using the simulated values, the hydraulic resistance was found to be **$5.09 \times 10^{10} \text{ Pa}\cdot\text{s}/\text{m}^3$** .

To quantify the difference between the analytical and numerical results, the percentage error was calculated as:

$$\% \text{Error} = \frac{|R_{\text{numerical}} - R_{\text{analytical}}|}{R_{\text{analytical}}} \times 100$$

The resulting error was approximately **13.87%**, which falls within an acceptable range. This discrepancy can be attributed to assumptions in the analytical model, including uniform velocity profiles and negligible entrance effects, whereas the numerical simulation captures more detailed flow dynamics. Furthermore, numerical approximations in ANSYS, such as meshing and boundary condition implementations, can contribute to minor variations.

5.5 Experimental Observations

Trial 1: Yellow and Blue Dyes

In the initial trial, a yellow and blue dye combination was used to visualize the concentration gradient. However, the yellow dye had low visibility, making it difficult to accurately observe the formation of the gradient. To improve contrast and clarity, the dye combination was modified to red and blue, which provided better differentiation and enhanced gradient visualization.

Trial 2: Red and Blue Dyes with Outlet Tubing

The second trial introduced outlet tubing to the setup, aiming to improve flow control. However, this led to leakage and instability due to pressure buildup within the tubing. The restricted outflow caused unintended mixing, disrupting the formation of a well-defined gradient. This trial highlighted the need for an optimized outlet design to maintain stability while preventing unwanted pressure accumulation.

Trial 3: No Outlet Tubing for Stability

To address the instability issues observed in Trial 2, the third trial was conducted without outlet tubing. This modification allowed for a more stable flow, enabling controlled diffusion-driven gradient formation. However, at the experimental flow rate of 0.2 mL/min, advection effects became dominant, deviating from the simulated flow conditions of 5.4×10^{-5} mL/min. This significant difference in flow rates altered the expected diffusion dynamics, emphasizing the importance of precise flow rate control for achieving the desired gradient profile.

6. Conclusions

The microfluidic gradient generator successfully demonstrated initial diffusion-driven mixing, with a distinct gradient formation observed under experimental conditions. However, deviations from the simulated flow rate resulted in advection-dominated mixing, highlighting the critical need for precise flow rate control to maintain stability. The lack of a dedicated waste outlet led to partial functionality, as minor leakage and unintended mixing affected overall performance. These limitations indicate that further refinements are necessary to ensure the system operates under optimal conditions.

To enhance the design, integrating a waste outlet can help regulate outflow and prevent backflow, thereby ensuring controlled fluid dynamics. Additionally, refining the channel geometry can enhance the stability of the concentration gradient, reducing undesired disturbances in the flow. Increasing the channel depth can further improve syringe connector compatibility and enhance the overall robustness of the system, making it more adaptable for various experimental conditions. Implementing these refinements will lead to more reliable, repeatable, and precise experimental outcomes, ensuring the microfluidic device performs as intended across different flow conditions.

7. Future Scope

- **Machine Learning for Design Optimization:** AI-driven simulations can be employed to predict and refine gradient profiles, enhancing the precision and adaptability of the microfluidic system for various applications.
- **Biological Applications:** The device can be tested with biomolecules to explore its potential in drug delivery and tissue engineering, providing insights into controlled biochemical environments.
- **Microfluidic Integration:** Future developments may incorporate real-time monitoring through sensor integration, allowing for precise control and analysis of gradient formation.
- **Optimization of Flow Conditions:** Adjustments in flow rates and channel dimensions can further improve gradient stability, ensuring more consistent and reproducible experimental conditions.

References

1. **Wang, X., Liu, Z., and Pang, Y., 2017.** Concentration gradient generation methods based on microfluidic systems. *RSC Advances*, **7**, pp.29966-29984.
2. **Pattanayak, P., Singh, S.K., Gulati, M., Vishwas, S., Kapoor, B., Chellappan, D.K., Anand, K., Gupta, G., Jha, N.K., Gupta, P.K., Prasher, P., Dua, K., Dureja, H., Kumar, D., and Kumar, V., 2021.** Microfluidic chips: recent advances, critical strategies in design, applications, and future perspectives. *Microfluidics and Nanofluidics*, **25**, article number 99.